one another by means of forces; and there is a universal force: gravity – every body attracts every other body.

An immense vision takes shape. Suddenly, after millennia, there is no longer separation between the heavens and Earth; there is no 'natural level' for things, as Aristotle presumed; there is no centre to the world; things let free no longer seek their natural place but move in a straight line, for ever.

A simple calculation with the little moon allows Newton to deduce how the force of gravity changes with distance and to determine its strength,* given by what we call today Newton's constant, indicated by the letter 'G' for 'gravity'. On Earth, this force causes things to fall; in the heavens it holds planets and satellites on their orbits. The force is the same.

It's a subversion of the conceptual structure of the Aristotelian world, the dominant vision of the world throughout the Middle Ages. Think of Dante's universe, for example: just as for Aristotle, Earth is a ball at the centre of the universe, orbited by the celestial spheres. Not any more. The universe is an immense, infinite space studded with stars, without limit and without centre. Within it, material bodies run free and straight, unless a force, generated by another body, deviates them. Reference to ancient atomism is clear in Newton, even if still formulated in conventional terms:

> It seems probable to me that God, in the beginning, formed matter in solid, massy, hard, impenetrable, moveable particles, of such sizes and figures, and with such other properties, and in such proportions to space …[3]

The world of Newtonian mechanics is simple, and is summarized in figures 2.1 and 2.2. It is the world of Democritus reborn. A world made of a vast, undifferentiated space, always equal to itself, where particles run for ever and act upon each other – and nothing else. The world sung by Leopardi:

> … sitting here and gazing, I find that endless
> Spaces beyond that hedge, and more-than-human
> Silences, and the deepest peace and quiet
> Are fashioned in my thought …[4]

But now the vision is immensely more powerful than in Democritus, because it is not just a mental image with which to order the world: it is now combined with mathematics, with the legacy of Pythagoras, and with the mighty tradition of the Alexandrian astronomers' mathematical physics. The world of Newton is the world of Democritus, rendered mathematical.

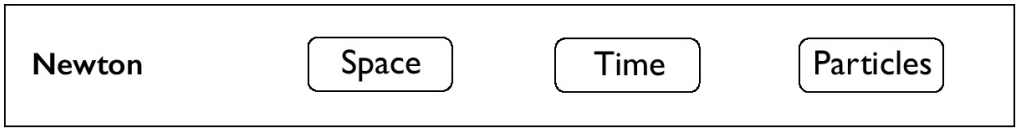


Figure 2.1 What is the world made of?